

Emission Factors and Reference Values

Version 3.0
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Canada's Greenhouse Gas
Offset Credit System



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Document revision history

Version number	Publication Date	Summary of changes
3.0	October 24, 2025	Addition of emission factors and other reference values applicable to the <i>Reducing Enteric Methane Emissions from Beef Cattle</i> federal offset protocol Update of emission factors and other reference values in alignment with the <i>National Inventory Report 1990–2023: Greenhouse Gas Sources and Sinks in Canada</i>
2.0	May 6, 2024	Addition of reference values applicable to the <i>Improved Forest Management on Private Land</i> federal offset protocol Update of emission factors and other reference values in alignment with the <i>National Inventory Report 1990–2022: Greenhouse Gas Sources and Sinks in Canada</i> Addition of rules on the timing applicability of emission factors and other reference values Changes to clarify the structure and facilitate the use of the document
1.1	June 13, 2023	Update of information and emission factors in alignment with the April 2023 publication of the <i>National Inventory Report 1990–2021: Greenhouse Gas Sources and Sinks in Canada</i>
1.0	June 8, 2022	Initial version

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1.0 Introduction

[Canada's Greenhouse Gas \(GHG\) Offset Credit System](#) is established under Part 2 of the *Greenhouse Gas Pollution Pricing Act* to provide an incentive to implement projects that result in domestic GHG reductions that would not have been generated in the absence of the project, that go beyond legal requirements and that are not subject to carbon pollution pricing mechanisms.

Canada's GHG Offset Credit System consists of:

- the [Canadian Greenhouse Gas Offset Credit System Regulations](#) (the Regulations), which establish the system, implement the operational aspects and set the general requirements applicable to all project types;
- federal offset protocols, included in the [Compendium of Federal Offset Protocols](#) (the Compendium), each containing requirements for project implementation and methods for quantifying GHG reductions for a given project type; and
- the [Credit and Tracking System](#) (CATS) to register offset projects, issue and track offset credits, and share key information through [Canada's GHG Offset Credit System Public Registry](#).

The Regulations apply to a proponent of a project which is of a type for which a protocol has been included in the Compendium; that aims to generate GHG reductions by preventing GHG emissions or removing GHGs from the atmosphere; and with respect to which the GHG reductions are real, additional, quantified, verified, unique and permanent. Offset credits will be issued to a proponent of a project for the period covered by a project report in the amount determined in accordance with subsection 29(2) of the Regulations if requirements of subsection 29(1) of the Regulations are met.

As per subsection 19(1) of the Regulations, this document provides emission factors and other reference values that a proponent must use in conjunction with a federal offset protocol to quantify the GHG reductions generated by a project. It also specifies timing applicability for the emission factors and other reference values that are to be used for the quantification of GHG reductions occurring in a given calendar year.

This document is categorized into general emission factors and other reference values that are applicable to more than one federal offset protocol, and protocol-specific emission factors and other reference values. For all emission factors and other reference values, it is specified which parameter in a given protocol the emission factors or other reference values correspond to. The proponent may need to convert the units of the emission factors and other reference values provided in this document to align with the units presented in the quantification methodology of the relevant federal offset protocol.

Emission factors and other reference values are subject to periodic updates when a new federal offset protocol is included in the Compendium, or when updated versions of the sources referenced in this document are published. As per subsection 1(2) of the Regulations, proponents must use the latest version of this document.

2.0 Abbreviations and acronyms

The following abbreviations and acronyms are used in this document:

Act	<i>Greenhouse Gas Pollution Pricing Act</i>
CH ₄	methane
CO ₂	carbon dioxide
CO ₂ e	carbon dioxide equivalent
g	gram
kg	kilogram
kWh	kilowatt hour
L	litre
m ³	cubic meter
N ₂ O	nitrous oxide
SF ₆	sulfur hexafluoride
t	metric tonne

Emission factors and other reference values contained in this document are often sourced from the annual [National inventory reports for GHG sources and sinks in Canada](#) (NIR), published by Environment and Climate Change Canada (ECCC). To simplify references to these reports in the footnotes, a short form citation is used. The complete citation of “Environment and Climate Change Canada. (year x). *National Inventory Report 1990–year y: Greenhouse Gas Sources and Sinks in Canada*” is replaced by “ECCC. (year x). NIR 1990–year y.” The short form citation in the footnotes also includes the relevant part number of the report, table number and table title corresponding to the emission factor or other reference value.

3.0 General

3.1 Global warming potentials

Global warming potentials to be used for the quantification of GHG reductions are the ones published in Schedule 3 to the Act at the time GHG reductions occur.

4.0 General emission factors and other reference values

Emission factors and other reference values in Section 4.0 may be applicable to more than one federal offset protocol.

4.1 Fossil fuel combustion

Emission factors contained in Tables 1.1, 1.2, and 1.3 correspond to the parameter EF_{CO_2} in applicable protocols.

Emission factors contained in Table 1.1 are to be used for the quantification of GHG reductions occurring in calendar years 2023 and 2024.

Table 1.1: CO₂ emission factors for natural gas (g CO₂/m³ natural gas) for 2023 and 2024

Province / Territory	Marketable ^{1*}	Non-marketable ^{2**}
British Columbia	1 966	2 162
Alberta	1 962	2 109
Saskatchewan	1 920	2 441
Manitoba	1 915	2 401
Ontario	1 921	2 401
Quebec	1 926	-
New Brunswick	1 919	2 401
Nova Scotia	1 919	2 494
Prince Edward Island	1 919	-
Newfoundland and Labrador	1 919	2 202
Yukon	1 966	2 401
Northwest Territories	1 966	2 466
Nunavut	1 966	-

* The term “marketable” applies to the fuel consumed by the Utility, Industry, Residential, Commercial, and Transport subsectors.

** The term “non-marketable” applies to raw/unprocessed gas consumption, mainly by natural gas producers.

Emission factors contained in Table 1.2 are to be used for the quantification of GHG reductions occurring in calendar year 2025.

Table 1.2: CO₂ emission factors for natural gas (g CO₂/m³ natural gas) for 2025

Province / Territory	Marketable ³	Non-marketable ⁴
British Columbia	1 966	2 162

¹ ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-1, “CO₂ Emission Factors for Marketable Natural Gas”

² ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-2, “CO₂ Emission Factors for Non-Marketable Natural Gas”

³ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-1, “CO₂ Emission Factors for Marketable Natural Gas”

⁴ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-2, “CO₂ Emission Factors for Non-Marketable Natural Gas”

Province / Territory	Marketable ³	Non-marketable ⁴
Alberta	1 962	2 113
Saskatchewan	1 920	2 441
Manitoba	1 915	2 401
Ontario	1 921	2 401
Quebec	1 926	-
New Brunswick	1 919	2 401
Nova Scotia	1 919	2 494
Prince Edward Island	1 919	-
Newfoundland and Labrador	1 919	2 340
Yukon	1 966	2 401
Northwest Territories	1 966	2 466
Nunavut	1 966	-

Emission factors contained in Table 1.3 are to be used for the quantification of GHG reductions occurring in calendar year 2026.

Table 1.3: CO₂ emission factors for natural gas (g CO₂/m³ natural gas) for 2026

Province / Territory	Marketable ⁵	Non-marketable ⁶
British Columbia	1 966	2 162
Alberta	1 962	2 112
Saskatchewan	1 920	2 441
Manitoba	1 915	2 401
Ontario	1 921	2 401
Quebec	1 926	-
New Brunswick	1 919	2 401
Nova Scotia	1 919	2 494
Prince Edward Island	1 919	-
Newfoundland and Labrador	1 919	2 260
Yukon	1 966	2 401

⁵ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.1-1, "CO₂ Emission Factors for Marketable Natural Gas"

⁶ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.1-2, "CO₂ Emission Factors for Non-Marketable Natural Gas"

Province / Territory	Marketable ⁵	Non-marketable ⁶
Northwest Territories	1 966	2 466
Nunavut	1 966	-

Emission factors contained in Tables 2.1, 2.2, and 2.3 correspond to the parameters EF_{CH_4} or EF_{N_2O} in applicable protocols.

Emission factors contained in Table 2.1 are to be used for the quantification of GHG reductions occurring in calendar years 2023 and 2024.

Table 2.1: CH₄ and N₂O emission factors for natural gas (g GHG/m³ natural gas)⁷ for 2023 and 2024

Source	CH ₄	N ₂ O
Electric Utilities	0.490	0.049
Industrial	0.037	0.033
Producer Consumption (Non-marketable)	6.4	0.060
Producer Consumption (Non-marketable) – Newfoundland and Labrador	0.490	0.060
Pipelines	1.900	0.050
Cement	0.037	0.034
Manufacturing Industries	0.037	0.033
Residential, Construction, Commercial/Institutional, Agriculture	0.037	0.035

Emission factors contained in Table 2.2 are to be used for the quantification of GHG reductions occurring in calendar year 2025.

Table 2.2: CH₄ and N₂O emission factors for natural gas (g GHG/m³ natural gas)⁸ for 2025

Source	CH ₄	N ₂ O
Electric Utilities	0.490	0.049
Industrial	0.037	0.033
Producer Consumption (Non-marketable) – Newfoundland and Labrador	0.490	0.060

⁷ ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-3, "CH₄ and N₂O Emission Factors for Natural Gas"

⁸ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-3, "CH₄ and N₂O Emission Factors for Natural Gas"

Source	CH ₄	N ₂ O
Producer Consumption (Non-marketable) – British Columbia	8.83 ⁹	0.060
Producer Consumption (Non-marketable) – Alberta	8.22 ⁹	0.060
Producer Consumption (Non-marketable) – Saskatchewan	4.64 ⁹	0.060
Producer Consumption (Non-marketable) – Other provinces and territories	6.4	0.060
Pipelines	1.900	0.050
Cement	0.037	0.034
Manufacturing Industries	0.037	0.033
Residential, Construction, Commercial/Institutional, Agriculture	0.037	0.035

Emission factors contained in Table 2.3 are to be used for the quantification of GHG reductions occurring in calendar year 2026.

Table 2.3: CH₄ and N₂O emission factors for natural gas (g GHG/m³ natural gas)¹⁰ for 2026

Source	CH ₄	N ₂ O
Electric Utilities	0.490	0.049
Industrial	0.037	0.033
Producer Consumption (Non-marketable) – Newfoundland and Labrador	0.490	0.060
Producer Consumption (Non-marketable) – British Columbia	9.20 ¹¹	0.060
Producer Consumption (Non-marketable) – Alberta	8.22 ¹¹	0.060
Producer Consumption (Non-marketable) – Saskatchewan	4.20 ¹¹	0.060
Producer Consumption (Non-marketable) –	6.4	0.060

⁹ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-4, “CH₄ Emission Factors for Natural Gas Producer Consumption in Western Provinces”

¹⁰ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.1-3, “CH₄ and N₂O Emission Factors for Natural Gas”

¹¹ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.1-4, “CH₄ Emission Factors for Natural Gas Producer Consumption in Western Provinces”

Source	CH ₄	N ₂ O
Other provinces and territories		
Pipelines	1.900	0.050
Cement	0.037	0.034
Manufacturing Industries	0.037	0.033
Residential, Construction, Commercial/Institutional, Agriculture	0.037	0.035

Emission factors contained in Tables 3.1, 3.2, and 3.3 correspond to the parameters EF_{CO₂}, EF_{CH₄} or EF_{N₂O} in applicable protocols.

Emission factors contained in Table 3.1 are to be used for the quantification of GHG reductions occurring in calendar years 2023 and 2024.

Table 3.1: Emission factors for natural gas liquids (g GHG/L fuel)¹² for 2023 and 2024

Fuel	CO ₂	CH ₄	N ₂ O
Propane - Residential	1 515	0.027	0.108
Propane - All Other Uses	1 515	0.024	0.108
Ethane	986	0.024	0.108
Butane	1 747	0.024	0.108

Emission factors contained in Table 3.2 are to be used for the quantification of GHG reductions occurring in calendar year 2025.

Table 3.2: Emission factors for natural gas liquids (g GHG/L fuel)¹³ for 2025

Fuel	CO ₂	CH ₄	N ₂ O
Propane - Residential	1 515	0.027	0.108
Propane - All Other Uses	1 515	0.024	0.108
Ethane	986	0.024	0.108
Butane	1 747	0.024	0.108

Emission factors contained in Table 3.3 are to be used for the quantification of GHG reductions occurring in calendar year 2026.

¹² ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-4, "Emission Factors for Natural Gas Liquids"

¹³ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-5, "Emission Factors for Natural Gas Liquids"

Table 3.3: Emission factors for natural gas liquids (g GHG/L fuel)¹⁴ for 2026

Fuel	CO ₂	CH ₄	N ₂ O
Propane - Residential	1 515	0.027	0.108
Propane - All Other Uses	1 515	0.024	0.108
Ethane	986	0.024	0.108
Butane	1 747	0.024	0.108

Emission factors contained in Tables 4.1, 4.2, and 4.3 correspond to the parameters EF_{CO₂}, EF_{CH₄} or EF_{N₂O} in applicable protocols.

Emission factors contained in Table 4.1 are to be used for the quantification of GHG reductions occurring in calendar years 2023 and 2024.

Table 4.1: Emission factors for refined petroleum products (g GHG/L fuel)¹⁵ for 2023 and 2024

Fuel	CO ₂	CH ₄	N ₂ O
Light Fuel Oil - Electric Utilities	2 753	0.18	0.031
Light Fuel Oil - Industrial	2 753	0.006	0.031
Light Fuel Oil - Producer Consumption	2 670	0.006	0.031
Light Fuel Oil - Residential	2 753	0.026	0.006
Light Fuel Oil - Forestry, Construction, Public Administration and Commercial/Institutional	2 753	0.026	0.031
Heavy Fuel Oil - Electric Utilities	3 156	0.034	0.064
Heavy Fuel Oil - Industrial	3 156	0.12	0.064
Heavy Fuel Oil - Producer Consumption	3 190	0.12	0.064
Heavy Fuel Oil - Residential, Forestry, Construction, Public Administration and Commercial/Institutional	3 156	0.057	0.064
Kerosene - Electric Utilities	2 560	0.006	0.031
Kerosene - Industrial	2 560	0.006	0.031
Kerosene - Producer Consumption	2 560	0.006	0.031
Kerosene - Residential	2 560	0.026	0.006

¹⁴ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.1-5, "Emission Factors for Natural Gas Liquids"

¹⁵ ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-5, "Emission Factors for Refined Petroleum Products"

Fuel	CO ₂	CH ₄	N ₂ O
Kerosene - Forestry, Construction, Public Administration and Commercial/Institutional	2 560	0.026	0.031
Diesel - Refineries and Others	2 681	0.078	0.022
Diesel - Upgraders	2 681	0.078	0.022
Petroleum Coke - Refineries and Others	3 877 ¹⁶	0.12	27.5 g/m ³ ¹⁷
Petroleum Coke - Upgraders	3 494 ¹⁶	0.12	24.0 g/m ³ ¹⁷
Still Gas - Refineries and Others	1 755 g/m ³ ¹⁶	0.032 g/m ³ ¹⁸	0.00002
Still Gas - Upgraders	2 140 g/m ³ ¹⁶	0.000039	0.00002
Motor Gasoline	2 307	0.100	0.02

Emission factors contained in Table 4.2 are to be used for the quantification of GHG reductions occurring in calendar year 2025.

Table 4.2: Emission factors for refined petroleum products (g GHG/L fuel)¹⁹ for 2025

Fuel	CO ₂	CH ₄	N ₂ O
Light Fuel Oil - Electric Utilities	2 753	0.18	0.031
Light Fuel Oil - Industrial	2 753	0.006	0.031
Light Fuel Oil - Producer Consumption	2 670	0.006	0.031
Light Fuel Oil - Residential	2 753	0.026	0.006
Light Fuel Oil - Forestry, Construction, Public Administration and Commercial/Institutional	2 753	0.026	0.031
Heavy Fuel Oil - Electric Utilities	3 156	0.034	0.064
Heavy Fuel Oil - Industrial	3 156	0.12	0.064
Heavy Fuel Oil - Producer Consumption	3 190	0.12	0.064
Heavy Fuel Oil - Residential, Forestry, Construction, Public Administration and Commercial/Institutional	3 156	0.057	0.064
Kerosene - Electric Utilities	2 560	0.006	0.031

¹⁶ ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-6, "CO₂ Emission Factors for Petroleum Coke and Still Gas"

¹⁷ ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-7, "N₂O Emission Factors for Petroleum Coke"

¹⁸ ECCC. (2023). NIR 1990-2021, Part 2, Table A6.1-8, "CH₄ Emission Factors for Still Gas (Refineries and Others)"

¹⁹ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-6, "Emission Factors for Refined Petroleum Products"

Fuel	CO ₂	CH ₄	N ₂ O
Kerosene - Industrial	2 560	0.006	0.031
Kerosene - Producer Consumption	2 560	0.006	0.031
Kerosene - Residential	2 560	0.026	0.006
Kerosene - Forestry, Construction, Public Administration and Commercial/Institutional	2 560	0.026	0.031
Diesel - Refineries and Others	2 681	0.078	0.022
Diesel - Upgraders	2 681	0.078	0.022
Petroleum Coke - Refineries and Others	3 776 ²⁰	0.12	27.5 g/m ³ ²¹
Petroleum Coke - Upgraders	3 494 ²⁰	0.12	24.0 g/m ³ ²¹
Still Gas - Refineries and Others	1 780 g/m ³ ²⁰	0.032 g/m ³ ²²	0.00002
Still Gas - Upgraders	2 140 g/m ³ ²⁰	0.000039	0.00002
Motor Gasoline	2 307	0.100	0.02

Emission factors contained in Table 4.3 are to be used for the quantification of GHG reductions occurring in calendar year 2026.

Table 4.3: Emission factors for refined petroleum products (g GHG/L fuel)²³ for 2026

Fuel	CO ₂	CH ₄	N ₂ O
Light Fuel Oil - Electric Utilities	2 753	0.18	0.031
Light Fuel Oil - Industrial	2 753	0.006	0.031
Light Fuel Oil - Producer Consumption	2 670	0.006	0.031
Light Fuel Oil - Residential	2 753	0.026	0.006
Light Fuel Oil - Forestry, Construction, Public Administration and Commercial/Institutional	2 753	0.026	0.031
Heavy Fuel Oil - Electric Utilities	3 156	0.034	0.064
Heavy Fuel Oil - Industrial	3 156	0.12	0.064
Heavy Fuel Oil - Producer Consumption	3 190	0.12	0.064

²⁰ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-7, "CO₂ Emission Factors for Petroleum Coke and Still Gas"

²¹ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-8, "N₂O Emission Factors for Petroleum Coke"

²² ECCC. (2024). NIR 1990-2022, Part 2, Table A6.1-9, "CH₄ Emission Factors for Still Gas (Refineries and Others)"

²³ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.1-6, "Emission Factors for Refined Petroleum Products"

Fuel	CO ₂	CH ₄	N ₂ O
Heavy Fuel Oil - Residential, Forestry, Construction, Public Administration and Commercial/Institutional	3 156	0.057	0.064
Kerosene - Electric Utilities	2 560	0.006	0.031
Kerosene - Industrial	2 560	0.006	0.031
Kerosene - Producer Consumption	2 560	0.006	0.031
Kerosene - Residential	2 560	0.026	0.006
Kerosene - Forestry, Construction, Public Administration and Commercial/Institutional	2 560	0.026	0.031
Diesel - Refineries and Others	2 681	0.078	0.022
Diesel - Upgraders	2 681	0.078	0.022
Petroleum Coke - Refineries and Others	3 254 ²⁴	0.12	23.6 g/m ³ ²⁵
Petroleum Coke - Upgraders	2 717 ²⁴	0.12	18.7 g/m ³ ²⁵
Still Gas - Refineries and Others	1 773 g/m ³ ²⁴	0.032 g/m ³ ²⁶	0.00002
Still Gas - Upgraders	2 140 g/m ³ ²⁴	0.000039	0.00002
Motor Gasoline	2 307	0.100	0.02

4.2 Grid electricity consumption

A 'consumption intensity' indicator is derived to reflect the GHG emissions intensity of electricity as it is delivered to the consumer.

Emission factors contained in Tables 5.1, 5.2, and 5.3 correspond to the parameter EF_{EL,GHG} in applicable protocols.

Emission factors contained in Table 5.1 are to be used for the quantification of GHG reductions occurring in calendar years 2023 and 2024.

²⁴ ECCC. (2025). NIR 1990-2023. Part 2, Table A6.1-7, "CO₂ Emission Factors for Petroleum Coke and Still Gas"

²⁵ ECCC. (2025). NIR 1990-2023. Part 2, Table A6.1-8, "N₂O Emission Factors for Petroleum Coke"

²⁶ ECCC. (2025). NIR 1990-2023. Part 2, Table A6.1-9, "CH₄ Emission Factors for Still Gas (Refineries and Others)"

Table 5.1: Electricity consumption intensities (g CO₂e/kWh electricity consumed)²⁷ for 2023 and 2024

Province / Territory	Consumption intensity ²⁸
British Columbia	15
Alberta	540
Saskatchewan	730
Manitoba	2.0
Ontario	30
Quebec	1.7
New Brunswick	300
Nova Scotia	690
Prince Edward Island ²⁹	300
Newfoundland and Labrador	17
Yukon	80
Northwest Territories	170
Nunavut	840

Emission factors contained in Table 5.2 are to be used for the quantification of GHG reductions occurring in calendar year 2025.

Table 5.2: Electricity consumption intensities (g CO₂e/kWh electricity consumed)³⁰ for 2025

Province / Territory	Consumption intensity ³¹
British Columbia	15
Alberta	490
Saskatchewan	670
Manitoba	1.4
Ontario	38
Quebec	1.7

²⁷ ECCC. (2023). NIR 1990-2021, Part 3, Tables A13-2 to A13-14

²⁸ Consumption intensity values are impacted by unallocated energy and SF₆ transmission emissions.

²⁹ Due to the high level of imports from New Brunswick, Prince Edward Island takes New Brunswick's value.

³⁰ ECCC. (2024). NIR 1990-2022, Part 3, Tables A13-2 to A13-14

³¹ Consumption intensity values are impacted by unallocated energy and SF₆ transmission emissions.

Province / Territory	Consumption intensity ³¹
New Brunswick	350
Nova Scotia	700
Prince Edward Island ³²	350
Newfoundland and Labrador	18
Yukon	70
Northwest Territories	190
Nunavut	820

Emission factors contained in Table 5.3 are to be used for the quantification of GHG reductions occurring in calendar year 2026.

Table 5.3: Electricity consumption intensities (g CO₂e/kWh electricity consumed)³³ for 2026

Province / Territory	Consumption intensity ³⁴
British Columbia	18
Alberta	438
Saskatchewan	631
Manitoba	2.5
Ontario	59
Quebec	1.9
New Brunswick	234
Nova Scotia	581
Prince Edward Island ³⁵	234
Newfoundland and Labrador	17
Yukon	74
Northwest Territories	420
Nunavut	800

³² Due to the high level of imports from New Brunswick, Prince Edward Island takes New Brunswick's value.

³³ ECCC. (2025). NIR 1990-2023, Part 3, Tables A13-2 to A13-14

³⁴ Consumption intensity values are impacted by unallocated energy and SF₆ transmission emissions.

³⁵ Due to the high level of imports from New Brunswick, Prince Edward Island takes New Brunswick's value.

4.3 Biogas combustion

Biogas includes landfill gas.

Emission factors contained in Tables 6.1, 6.2, and 6.3 correspond to the parameter EF_{LFG,N_2O} in applicable protocols.

Emission factors contained in Table 6.1 are to be used for the quantification of GHG reductions occurring in calendar years 2023 and 2024.

Table 6.1: N_2O emission factors for biogas combustion ($kg\ N_2O/t\ CH_4$)³⁶ for 2023 and 2024

Description	N_2O
Combustion of biogas for energy through a boiler, turbine, internal combustion engine or station for natural gas network	0.005
Flaring of biogas ³⁷	0

Emission factors contained in Table 6.2 are to be used for the quantification of GHG reductions occurring in calendar year 2025.

Table 6.2: N_2O emission factors for biogas combustion ($kg\ N_2O/t\ CH_4$)³⁸ for 2025

Description	N_2O
Combustion of biogas for energy through a boiler, turbine, internal combustion engine or station for natural gas network	0.005
Flaring of biogas ³⁹	0.005

Emission factors contained in Table 6.3 are to be used for the quantification of GHG reductions occurring in calendar year 2026.

Table 6.3: N_2O emission factors for biogas combustion ($kg\ N_2O/t\ CH_4$)⁴⁰ for 2026

Description	N_2O
Combustion of biogas for energy through a boiler, turbine, internal combustion engine or station for natural gas network	0.005
Flaring of biogas ³⁹	0.005

³⁶ ECCC. (2023). NIR 1990-2021, Part 2, Table A6.6-2, "Emission Factors for Landfill Gas Combustion"

³⁷ This emission factor is currently reported as "not estimated" in Canada's National Inventory Report. A value of zero has been assigned to this emission factor for the purposes of the quantification of GHG reductions.

³⁸ ECCC. (2024). NIR 1990-2022, Part 2, Table A6.6-2, "Emission Factors for Landfill Gas Combustion"

³⁹ This emission factor is currently reported as "not estimated" in Canada's National Inventory Report. To be conservative, the emission factor applicable to combustion of biogas for energy is applied to flaring of biogas.

⁴⁰ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.6-2, "Emission Factors for Landfill Gas Combustion"

5.0 Protocol-specific emission factors and other reference values

Emission factors and other reference values in Section 5.0 are only applicable to the federal offset protocol specified.

5.1 Improved forest management on private land

Reference values contained in Tables 7 and 8 are to be used for the quantification of GHG reductions generated by projects implemented following the *Improved Forest Management on Private Land* protocol and occurring from calendar year 2024 onwards.

Reference values contained in Table 7 correspond to the parameter $PC_{i,C}$ in the protocol.

Table 7: Percentage of harvest by wood product class⁴¹

Wood product class	Percentage of harvest (%)
Softwood lumber	37.76
Hardwood lumber	0.38
Pulp and paper	34.60
Panels (plywood and oriented strandboard)	12.42
Other industrial roundwood	3.55
Fuelwood	11.29

Reference values contained in Table 8 correspond to the parameter SF_j in the protocol.

Table 8: 100-year storage factor by wood product class⁴²

Wood product class	100-year storage factor
Softwood lumber	0.213
Hardwood lumber	0.156
Softwood plywood	0.215
Oriented strandboard	0.285
Non-structural panels	0.174

⁴¹ Values are based off the national average harvested wood product production ratios developed by the Food and Agriculture Organization (FAO) based on a reference period of 1990-2020. Ratio of softwood lumber to hardwood lumber for harvested industrial roundwood was based on a reference period of 2014-2021 from Statistics Canada (Table 16-10-0017-01 Lumber production, shipments, and stocks by species, monthly (x 1,000), [doi: 10.25318/1610001701-eng](https://doi.org/10.25318/1610001701-eng)).

⁴² Hoover et al. (2014). [Chapter 6: quantifying greenhouse gas sources and sinks in managed forest systems](#). M., D. Pape, M. Flugge, R. Steele, D. Man, M. Riley-Gilbert, and S. Biggar, (Eds.). *Greenhouse Gas Fluxes in Agriculture and Forestry: Methods for Entity-Scale Inventory* (Technical Bulletin Number 1939). US Department of Agriculture.

Wood product class	100-year storage factor
Other industrial roundwood	0.149
Fuelwood	0
Pulp and paper	0

5.2 Reducing enteric methane emissions from beef cattle

Emission factors and other reference values contained in Tables 9 to 12 are to be used for the quantification of GHG reductions generated by projects implemented following the *Reducing Enteric Methane Emissions from Beef Cattle* protocol and occurring from calendar year 2025 onwards.

Reference values contained in Table 9 correspond to the parameter Y_m in the protocol.

Table 9: Enteric CH₄ conversion factors based on diet composition and quality⁴³

Diet description	Enteric CH ₄ conversion factor (Y_m)
Diets of more than 75% low to medium quality forage containing < 60% total digestible nutrients	0.07
Diets of more than 75% high quality forage containing ≥ 60% total digestible nutrients	0.063
Mixed diets with forage content of 15 to 75% and the total diet is mixed with grain	0.063
All other grains with 0 to 15% forage	0.04
Steam-flaked corn and ionophore supplement with 0 to 10% forage	0.03

Emission factors contained in Table 10 correspond to the parameter EF_{lip} in the protocol.

Table 10: Emission factors for the addition of supplemented lipid determined by the percentage of total dry weight of feed delivered

Supplemented lipid added (%)	Emission factor (EF_{lip})
< 1.0	1.0
1.0 to 1.99	0.96

⁴³ Reference values from Table 9 are adapted from International Panel on Climate Change. (2019). Chapter 10: Agriculture, Forestry and Other Land Use. Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize S., Osako, A., Pyrozhenko, Y., Shermanau, P. and Federici, S. (eds). in [2019 refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories](#), Table 10.12.

Supplemented lipid added (%)	Emission factor (EF _{lip})
2.0 to 2.99	0.92
3.0 to 3.99	0.88
4.0 to 4.99	0.84
≥ 5.0	0.80

Emission factors and other reference values contained in Table 11 correspond to the parameters MCF, EF_{MS}, Frac_v, and Frac_L in the protocol.

Table 11: Emission factors and other reference values for manure CH₄ and N₂O based on manure storage system

Manure storage system ⁴⁴	CH ₄ conversion factor (MCF) ⁴⁵	Emission factor for direct N ₂ O emissions (EF _{MS}) ⁴⁶	Fraction of nitrogen (N) excreted in manure that volatilizes (Frac _v) ⁴⁷	Fraction of N excreted in manure leached (Frac _L) ⁴⁸
Solid storage and dry lot	0.02	0.02	0.3	0.03
Liquid, slurry or pit storage below confinements	0.2	0.001	0.4	0
Other manure storage system	0.01	0.005	0.24 ⁴⁸	0.05

⁴⁴ Description of each type of manure storage system can be found in International Panel on Climate Change. (2019). 2019 refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4, [Chapter 10 Emissions from Livestock and Manure Management](#), Table 10.18.

⁴⁵ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.4-4, "Methane Conversion Factors (MCF) by Animal Category and Manure Management System"

⁴⁶ ECCC. (2025). NIR 1990-2023, Part 2, Table A6.4-10, "Emission Factors (EF) for Manure Nitrogen (N) Lost as N₂O-N by Animal Category and Animal Waste Management Systems"

⁴⁷ ECCC. (2025). NIR 1990-2023, Part 2, Table A3.4-27, "Total N, NH₃- and NOx-N Losses Associated with Various Livestock and Manure Management Systems"

⁴⁸ ECCC. (2025). NIR 1990-2023, Part 2, Table A3.4-28, "Total N, NH₃- and NOx-N Losses Associated with Manure Management Systems for Dairy Cattle"

Emission factors contained in Table 12 correspond to the parameter EF_V in the protocol.

Table 12: Emission factors for indirect N_2O emissions from volatilization of manure by ecozone⁴⁹

Ecozone	Emission factor for indirect N_2O emissions from volatilization of manure (EF_V)
Taiga Plains	0.005
Boreal Shield	0.014
Atlantic Maritime	0.014
Mixedwood Plains	0.014
Boreal Plains	0.005
Prairies	0.005
Pacific Maritime	0.014
Montane Cordillera	0.005

⁴⁹ ECCC. (2025). NIR 1990-2023, Part 2, Table A3.4-26, "Application of Emission Factor EF_4 from the 2019 Refinement to the 2006 IPCC Guidelines, by Ecozone"